

Exercise Set 7

Quantum Computer Programming

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1. [IBMQE] Challenge problem: *“The Deutsch–Jozsa algorithm. Recall that you had a question above asking you to work out the intermediate qubit states π_1 and π_2 of the Deutsch’s algorithm. Do the same for the intermediate $(n+1)$ -qubit states π_1 and π_2 of the Deutsch–Jozsa algorithm, for the specific case $n = 2$. Then, verify that $\pi_3 = |-\rangle \otimes \frac{1}{2^n} \sum_{x \in \Sigma^n} \sum_{y \in \Sigma^n} (-1)^{(s \cdot x) + (x \cdot y)} |y\rangle$, again, for the specific case $n = 2$.”*

References

[IBMQE] Quantum Platform Courses: Deutsch-Jozsa Algorithm
(<https://quantum.cloud.ibm.com/learning/en/modules/computer-science/deutsch-jozsa>)